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Chapter 1

Easy way to memorize Div, grad, and
Curl

1.1 Einstein Summation Convention

Einstein (1916) introduced the following summation convention:

If an index appears exactly twice in a single term, it is understood that one should automatically sum over that index. Such an index is called a *dummy index*.

In this note, the indices i, j, k typically range from 1 to 3 in an orthogonal coordinate system, and we write:

$$\begin{aligned}\vec{V} &= \sum_{k=1}^3 V_k \hat{e}_k = V_k \hat{e}_k, & \vec{V} \cdot \vec{F} &= \sum_{k=1}^3 V_k F_k = V_k F_k \\ \nabla \varphi &= \hat{e}_k \partial_k \varphi = \hat{e}_k \frac{\partial \varphi}{\partial x_k}, & \nabla \cdot \vec{D} &= \partial_k D_k = \frac{\partial D_k}{\partial x_k} \\ C_{ik} &= \sum_{j=1}^M A_{ij} B_{jk} = A_{ij} B_{jk}\end{aligned}$$

Note that an index appearing only once is not a summation index.

1.2 Lamé coefficients in the orthogonal coordinates

Let $\vec{r}(u_1, u_2, u_3)$ define the position vector in a general orthogonal curvilinear system. There are three orthogonal coordinate systems of interest:

$$\text{Orthogonal coordinate systems } (u_1, u_2, u_3) \quad \begin{cases} \text{Cartesian :} & u_1 = x, & u_2 = y, & u_3 = z \\ \text{Cylindrical :} & u_1 = \rho, & u_2 = \phi, & u_3 = z \\ \text{Spherical :} & u_1 = r, & u_2 = \theta, & u_3 = \phi \end{cases}$$

Now, we define the **tangent vector** along the coordinate line as

$$\boxed{\vec{e}_i \equiv \frac{\partial \vec{r}}{\partial u_i}}$$

1. Cartesian coordinates (x, y, z) : $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\vec{e}_x = \frac{\partial \vec{r}}{\partial x} = \hat{i}, \quad \vec{e}_y = \frac{\partial \vec{r}}{\partial y} = \hat{j}, \quad \vec{e}_z = \frac{\partial \vec{r}}{\partial z} = \hat{k}$$

2. Cylindrical coordinates (ρ, ϕ, z) : $\vec{r} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z \hat{k}$

$$\vec{e}_r = \frac{\partial \vec{r}}{\partial \rho} = \cos \phi \hat{i} + \sin \phi \hat{j}, \quad \vec{e}_\phi = \frac{\partial \vec{r}}{\partial \phi} = -\rho \sin \phi \hat{i} + \rho \cos \phi \hat{j}, \quad \vec{e}_z = \frac{\partial \vec{r}}{\partial z} = \hat{k}$$

3. Spherical coordinates (r, θ, ϕ) : $\vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}$

$$\begin{aligned}\vec{e}_r &= \frac{\partial \vec{r}}{\partial r} = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k} \\ \vec{e}_\theta &= \frac{\partial \vec{r}}{\partial \theta} = r \cos \theta \cos \phi \hat{i} + r \cos \theta \sin \phi \hat{j} - r \sin \theta \hat{k} \\ \vec{e}_\phi &= \frac{\partial \vec{r}}{\partial \phi} = -r \sin \theta \sin \phi \hat{i} + r \sin \theta \cos \phi \hat{j}\end{aligned}$$

Note that these $\vec{e}_i$ s are not unit vectors. To normalize these tangent vectors, we define the magnitude of this tangent vector as *Lamé coefficient*(scale factor) h_i ,

$$h_i \equiv \|\vec{e}_i\|$$

Specifically, h_i s in each orthogonal coordinate systems are computed as follows:

$$\text{Lamé coefficients: } \begin{cases} \text{Cartesian } (x, y, z) : & h_x = 1, h_y = 1, h_z = 1 \\ \text{Cylindrical } (\rho, \phi, z) : & h_\rho = 1, h_\phi = \rho, h_z = 1, \\ \text{Spherical } (r, \theta, \phi) : & h_r = 1, h_\theta = r, h_\phi = r \sin \theta. \end{cases}$$

By dividing the tangent vectors by the Lamé coefficients, the unit orthogonal(*orthonormal*) basis corresponding to each coordinate direction are defined:

$$\hat{e}_i \equiv \frac{\vec{e}_i}{h_i}$$

1. Cartesian coordinates $(x, y, z) : \vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\vec{e}_i = \hat{e}_i \quad (\because h_i = 1)$$

2. Cylindrical coordinates $(\rho, \phi, z) : \vec{r} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z \hat{k}$

$$\begin{aligned} \hat{e}_r &= \vec{e}_r = \cos \phi \hat{i} + \sin \phi \hat{j} \\ \hat{e}_\phi &= \frac{\vec{e}_\phi}{r} = -\sin \phi \hat{i} + \cos \phi \hat{j} \\ \hat{e}_z &= \vec{e}_z = \hat{k} \end{aligned}$$

3. Spherical coordinates $(r, \theta, \phi) : \vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}$

$$\begin{aligned} \hat{e}_r &= \vec{e}_r = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k} \\ \hat{e}_\theta &= \frac{\vec{e}_\theta}{r} = \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} - \sin \theta \hat{k} \\ \hat{e}_\phi &= \frac{\vec{e}_\phi}{r \sin \theta} = -\sin \phi \hat{i} + \cos \phi \hat{j} \end{aligned}$$

Summary Table

Coordinate System	Tangential Vector $\vec{e}_i$	Magnitude h_i	Unit Vector $\hat{e}_i$
Cartesian (x, y, z)	$\hat{i}, \hat{j}, \hat{k}$	1	same
Cylindrical (r, ϕ, z)	$(\cos \phi, \sin \phi), (-r \sin \phi, r \cos \phi), (0, 0, 1)$	1, r , 1	$\vec{e}_i/h_i$
Spherical (r, θ, ϕ)	see above	1, r , $r \sin \theta$	$\vec{e}_i/h_i$

In summary, $\vec{e}_i$ is the actual *tangent vector* to the coordinate curve, h_i is the *Lamé coefficient* (*scale factor*), and $\hat{e}_i$ is the corresponding *normalized unit vector*. Only in the Cartesian system do we have $\vec{e}_i = \hat{e}_i$.

1.3 Memorizing Vector Derivative operators

The Lamé coefficients allow us to express vector differential operators in orthogonal curvilinear coordinate systems in a coordinate-free form.

As a result, it is unnecessary to memorize the explicit expressions of these operators in each individual coordinate system. Instead, by remembering only the coordinate-free formulation and the corresponding Lamé coefficients, one can easily derive the vector differential operators for any orthogonal coordinate system.

1.3.1 Gradient of a scalar field φ

(1) Coordinate-free form

$$\nabla\varphi = \hat{e}_i \frac{1}{h_i} \frac{\partial\varphi}{\partial u_i} \equiv \hat{e}_i \frac{\partial_i\varphi}{h_i}$$

(2) Common orthogonal coordinates

$$\text{Cartes. :} \quad h_x = h_y = h_z = 1, \quad \nabla\varphi = \hat{i} \partial_x\varphi + \hat{j} \partial_y\varphi + \hat{k} \partial_z\varphi,$$

$$\text{Cylind. :} \quad h_r = 1, h_\phi = r, h_z = 1, \quad \nabla\varphi = \hat{e}_r \partial_r\varphi + \hat{e}_\phi \frac{1}{r} \partial_\phi\varphi + \hat{e}_z \partial_z\varphi,$$

$$\text{Spheric. :} \quad h_r = 1, h_\theta = r, h_\phi = r \sin\theta, \quad \nabla\varphi = \hat{e}_r \partial_r\varphi + \hat{e}_\theta \frac{1}{r} \partial_\theta\varphi + \hat{e}_\phi \frac{1}{r \sin\theta} \partial_\phi\varphi.$$

1.3.2 Divergence of a Vector field $\vec{A}$

(1) Coordinate-free form

$$\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(\frac{h_1 h_2 h_3}{h_i} A_i \right) \equiv \frac{1}{h_1 h_2 h_3} \partial_i \left(\frac{h_1 h_2 h_3}{h_i} A_i \right)$$

(2) Common orthogonal coordinates

$$\text{Cartes. :} \quad h_x = h_y = h_z = 1, \quad \nabla \cdot \vec{A} = \partial_x A_x + \partial_y A_y + \partial_z A_z,$$

$$\text{Cylind. :} \quad h_r = 1, h_\phi = r, h_z = 1, \quad \nabla \cdot \vec{A} = \frac{1}{r} \partial_r (r A_r) + \frac{1}{r} \partial_\phi A_\phi + \partial_z A_z,$$

$$\text{Spheric. :} \quad h_r = 1, h_\theta = r, h_\phi = r \sin\theta, \quad \nabla \cdot \vec{A} = \frac{1}{r^2} \partial_r (r^2 A_r) + \frac{1}{r \sin\theta} \partial_\theta (\sin\theta A_\theta) + \frac{1}{r \sin\theta} \partial_\phi A_\phi.$$

1.3.3 Laplacian of a scalar field φ

(1) Coordinate-free form

$$\boxed{\nabla^2 \varphi = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \varphi}{\partial u_i} \right) \equiv \frac{1}{h_1 h_2 h_3} \partial_i \left(\frac{h_1 h_2 h_3}{h_i^2} \partial_i \varphi \right)}$$

(2) Common orthogonal coordinates

Cartes. : $h_x = h_y = h_z = 1, \quad \nabla^2 \varphi = \partial_x^2 \varphi + \partial_y^2 \varphi + \partial_z^2 \varphi,$

Cylind. : $h_r = 1, h_\phi = r, h_z = 1, \quad \nabla^2 \varphi = \frac{1}{r} \partial_r (r \partial_r \varphi) + \frac{1}{r^2} \partial_\phi^2 \varphi + \partial_z^2 \varphi,$

Spheric. : $h_r = 1, h_\theta = r, h_\phi = r \sin \theta, \quad \nabla^2 \varphi = \frac{1}{r^2} \partial_r (r^2 \partial_r \varphi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta \varphi) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \varphi.$

1.3.4 Curl of a vector field $\vec{A}$

(1) Coordinate-free form

$$\boxed{\nabla \times \vec{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_1 & h_2 \hat{e}_2 & h_3 \hat{e}_3 \\ \partial_1 & \partial_2 & \partial_3 \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}} \quad \partial_i \equiv \frac{\partial}{\partial u_i}.$$

(2) Cartesian coordinates (x, y, z)

: $h_x = h_y = h_z = 1$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ A_x & A_y & A_z \end{vmatrix}.$$

(3) Cylindrical coordinates (r, ϕ, z)

$h_r = 1, h_\phi = r, h_z = 1$

$$\nabla \times \vec{A} = \frac{1}{r} \begin{vmatrix} \hat{e}_r & r \hat{e}_\phi & \hat{e}_z \\ \partial_r & \partial_\phi & \partial_z \\ A_r & r A_\phi & A_z \end{vmatrix}.$$

(4) Spherical coordinates (r, θ, ϕ)

$h_r = 1, h_\theta = r, h_\phi = r \sin \theta$

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{e}_r & r \hat{e}_\theta & r \sin \theta \hat{e}_\phi \\ \partial_r & \partial_\theta & \partial_\phi \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}.$$

Warning. 자주하는 실수로 행렬식 전체에 곱해지는 상수 $\left(\frac{1}{h_1 h_2 h_3} \right)$ 를 깜박한다.

Chapter 2

Appendix. Derivation of coordinate-free vector differential operators

2.1 Gradient: Full Derivation

Goal: Derive an expression for $\nabla\phi$ in terms of Lamé coefficients.

We begin with the total differential of a scalar field:

$$d\phi = \sum_i \frac{\partial\phi}{\partial u^i} du^i \quad (2.1)$$

Now consider a small displacement in space:

$$d\vec{r} = \sum_i \vec{e}_i du^i = \sum_i h_i \hat{e}_i du^i \quad (2.2)$$

By definition of the gradient:

$$d\phi = \nabla\phi \cdot d\vec{r} \quad (2.3)$$

Substitute $d\vec{r}$:

$$\begin{aligned} d\phi &= \nabla\phi \cdot \sum_i h_i \hat{e}_i du^i \\ &= \sum_i h_i (\nabla\phi \cdot \hat{e}_i) du^i \end{aligned}$$

Matching both expressions for $d\phi$:

$$\frac{\partial\phi}{\partial u^i} = h_i (\nabla\phi \cdot \hat{e}_i) \Rightarrow \nabla\phi = \sum_i \frac{1}{h_i} \frac{\partial\phi}{\partial u^i} \hat{e}_i \quad (2.4)$$

2.2 Divergence: Full Derivation

Goal: Compute $\nabla \cdot \vec{A}$ for $\vec{A} = \sum_i A^i \hat{e}_i$.

We consider a small volume element:

$$dV = h_1 h_2 h_3 du^1 du^2 du^3 \quad (2.5)$$

From Gauss's theorem:

$$\nabla \cdot \vec{A} = \lim_{V \rightarrow 0} \frac{1}{V} \oint_{\partial V} \vec{A} \cdot d\vec{S}$$

For the i -th direction, the net flux is:

$$\left[\frac{\partial}{\partial u^i} (A^i h_j h_k) \right] du^1 du^2 du^3, \quad (i \neq j \neq k) \quad (2.6)$$

Then total divergence:

$$\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial u^i} (h_j h_k A^i) \quad (2.7)$$

Using $h_j h_k = \frac{h_1 h_2 h_3}{h_i}$:

$$\boxed{\nabla \cdot \vec{A} = \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial u^i} \left(h_1 h_2 h_3 \cdot \frac{A^i}{h_i} \right)} \quad (2.8)$$

2.3 Curl: Full Derivation

Goal: Derive $\nabla \times \vec{A}$ from Stokes' theorem.

Take the component $(\nabla \times \vec{A})^3$. Let surface lie in the u^1 - u^2 plane:

$$d\vec{S} = h_1 h_2 du^1 du^2 \hat{e}_3 \quad (2.9)$$

Circulation along the loop:

$$\oint \vec{A} \cdot d\vec{\ell} \approx \left(\frac{\partial A^1}{\partial u^2} h_1 - \frac{\partial A^2}{\partial u^1} h_2 \right) du^1 du^2 \quad (2.10)$$

Then:

$$(\nabla \times \vec{A})^3 = \frac{1}{h_1 h_2} \left(\frac{\partial}{\partial u^2} (h_1 A^1) - \frac{\partial}{\partial u^1} (h_2 A^2) \right) \quad (2.11)$$

Using cyclic permutation:

$$\boxed{\nabla \times \vec{A} = \sum_{\text{cyc}} \frac{1}{h_j h_k} \left[\frac{\partial}{\partial u^j} (h_k A^k) - \frac{\partial}{\partial u^k} (h_j A^j) \right] \hat{e}_i} \quad (2.12)$$

Levi-Civita Symbol Form

Using the identity:

$$(\nabla \times \vec{A})^i = \frac{1}{h_1 h_2 h_3} \sum_{j,k} \epsilon_{ijk} h_j \frac{\partial}{\partial u^j} (h_k A^k) \quad (2.13)$$

We get:

$$\boxed{\nabla \times \vec{A} = \frac{1}{h_1 h_2 h_3} \sum_{i,j,k} \epsilon_{ijk} h_j \frac{\partial}{\partial u^j} (h_k A^k) \hat{e}_i} \quad (2.14)$$

2.4 Laplacian: Full Derivation

Apply divergence to the gradient:

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \nabla \cdot \left(\sum_i \frac{1}{h_i} \frac{\partial \phi}{\partial u^i} \hat{e}_i \right) \quad (2.15)$$

$$= \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial u^i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \phi}{\partial u^i} \right) \quad (2.16)$$

$$\boxed{\nabla^2 \phi = \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial u^i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \phi}{\partial u^i} \right)} \quad (2.17)$$